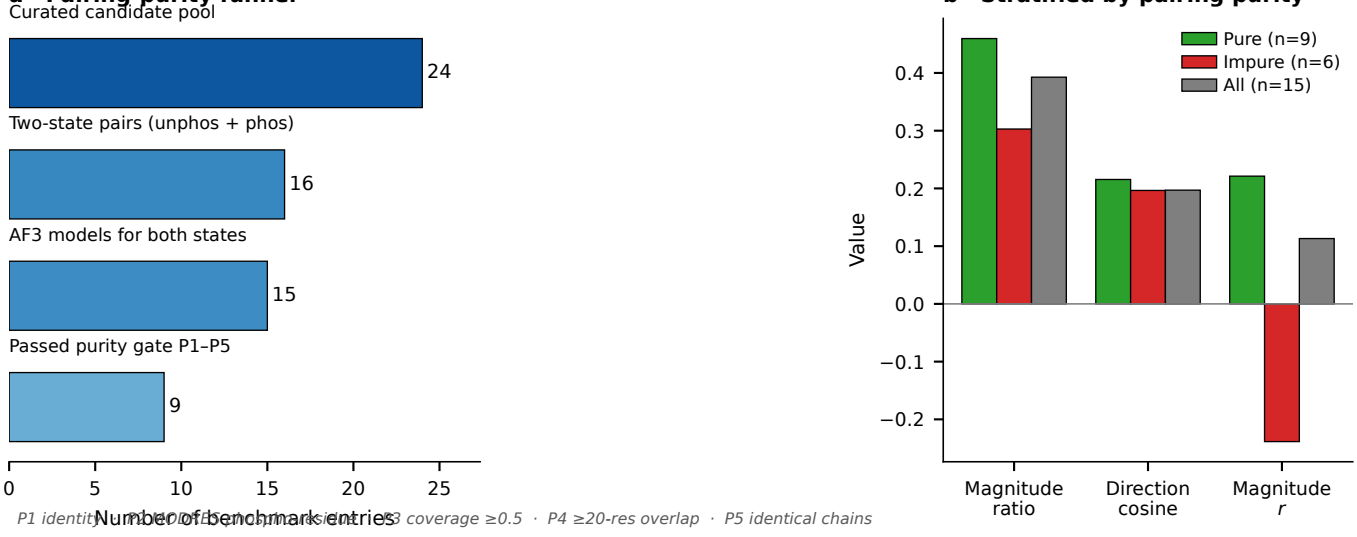
a Pairing-purity funnel
Curated candidate pool
24
Two-state pairs (unphos + phos)
16
AF3 models for both states
15
Passed purity gate P1–P5
9
0
5
10
15
20
25
Number of benchmark entries
b Stratified by pairing purity
Pure (n=9)
Impure (n=6)
All (n=15)
Value
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
Magnitude ratio
Direction cosine
Magnitude r
P1 identity-matched · P2 ≥2 chains · P3 ≥2 conditions · P4 coverage ≥0.5 · P5 ≥20-res overlap · P6 identical chains